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Energy, equity and inclusion: Exploring gender equality and social inclusion in energy planning in Kenya

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ABSTRACT

The provision of universal access to modern energy and the facilitation of a clean and equitable energy transition depends on the consideration of the needs of marginalised and vulnerable groups in energy planning. Kenya has made significant progress in increasing energy access and has implemented policies aimed at inclusivity and equality in energy and other sectors. However, questions remain about how gender equality and social inclusion (GESI) is understood, framed and operationalised by decision-makers across governance levels. Through a policy review and 23 semi-structured interviews with experts from national and county governments, civil society organisations and the private sector, this paper investigates how stakeholder perceptions towards GESI impact on broader structural framings in policy and planning in Kenya's energy sector. Findings reveal that while gender considerations are increasingly recognised, other forms of marginalisation and their intersectionality remain insufficiently addressed—some experts see GESI as only relating to gender, rather than something that includes a broader range of marginalising factors. Furthermore, the implementation of GESI-related commitments remains limited, particularly at the sub-national level, due to a lack of institutional capacity and resources. The paper concludes that Kenya has made notable strides to mainstream GESI in energy policy and planning; however, further efforts are required to dismantle structural barriers, including persistent gender norms and power imbalances.

1. Introduction

Universal access to affordable and clean energy is a key enabler for sustainable development, as highlighted in Sustainable Development Goal (SDG) 7 [1]. Energy fuels everyday activities such as cooking, working, travelling, and enjoying leisure activities [2,3]. However, in 2023, reportedly 666 million people globally had no access to electricity, and 2.1 billion people were relying on polluting fuels for cooking [4,5]. Over 80 % of individuals without electricity access live in sub-Saharan Africa [4]. The lack of access to energy typically impacts most profoundly people who have been marginalised [6], as they have specific and diverse requirements when it comes to accessing and using energy that are currently not accounted for [7]. Energy planning often happens in top-down approaches and focuses on facilitating access to energy without acknowledging diverse needs [8–13]. Therefore, it is vital that planning approaches include considerations of Gender Equality and Social Inclusion (GESI) to ensure that the benefits of access to energy are fully realised for all.

The concept of GESI aims to ensure that all groups of people – particularly those who are marginalised and vulnerable on the basis of characteristics such as gender, age, disability, ethnicity, location, or socio-economic status — have equitable access to opportunities, resources, and decision-making processes [14,15]. Utilising GESI as a framework to systematically examine who is included, who is excluded, and the extent of consideration given to the needs of different groups in policymaking processes allows researchers to assess distributional impacts and address structural inequalities.

The body of literature on GESI (rather than gender research more generally) at the interface with, and its application to, different sectors is nascent but growing. Some studies focus on gender or GESI in agriculture, agribusiness, water and hygiene [24–33], and there is a small, but growing, knowledge base on GESI in energy [34–36]. Two recent contributions have underscored the importance of integrating GESI into energy planning and modelling in low- and middle-income countries

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(LMICs) [9,10] highlighting how failure to account for social inclusion and gender dynamics in quantitative approaches can perpetuate structural inequalities. In energy, GESI approaches aim to explicitly address underlying structural barriers to energy access and use faced by different groups and to make energy transitions more equitable. To realise this, energy decision-makers need a good understanding of the diverse energy needs and challenges people face, and of the opportunities that inclusive planning can bring for marginalised and vulnerable groups. This involves assessing who benefits from energy measures, who does not, and why [16,17].

This research aims to understand the role of GESI in energy planning approaches by drawing insights from Kenya's experience. It focuses on Kenya because, while the country has made significant progress in mainstreaming gender in national planning and development frameworks - such as through commitments to equality in the Constitution of Kenya of 2010, the Vision 2030 agenda [18–20] - the broader application of GESI remains limited. Marginalised and vulnerable groups continue to face structural barriers to equitable energy access, including reliance on polluting biomass fuels, affordability constraints, geographic exclusion, and discrimination [8,21–23]. Given Kenya's devolution of energy planning, which mandates all 47 counties to develop County Energy Plans (CEPs), there is also a gap in understanding perceptions towards GESI of decision-makers across levels of governance [7,10,37]. The paper has the following research objectives, to: 1) understand how energy and gender experts understand and perceive GESI; 2) examine the extent to which GESI considerations are integrated in national and county-level energy policy and planning; and, 3) explore how the integration of GESI in Kenya's energy planning can be strengthened.

Drawing on a policy review and insights from 23 semi-structured interviews with energy-sector decision-makers in Kenya, this paper analyses stakeholder perceptions of GESI and how these may shape broader structural framing of policy and practice. Building on this analysis, the study identifies barriers that currently hinder greater inclusion of GESI in energy planning, as well as potential strategies to strengthen its integration into energy planning. In the East African region specifically, the number of studies on GESI in energy planning is limited and this analysis therefore also relies on case studies from other world regions. In doing so, it not only highlights key challenges and successful approaches relevant for decision-makers in Kenya but also offers lessons for other countries seeking to adopt more inclusive energy planning practices.

The paper begins with an introduction to GESI in energy planning, focusing on Kenya's evolving policy frameworks for gender and for energy, and outlines the intersection of the two sectors. This is followed by a description of the methods used in this research, which include a policy review and expert interviews. It then presents the analysis which explores how GESI is understood, examines the marginalisation of different social groups in energy planning, and discusses implementation challenges, before outlining participants' recommendations for improving GESI integration in energy planning. The paper concludes by suggesting that, while Kenya continues to be a world leader in delivering access to energy, and has made progress towards GESI, there is significant need in key areas for more be done.

2. GESI in energy planning

Gender equality and social inclusion are human rights, and a vital requirement for equitable societal development [14]. With its focus on both gender equality and social inclusion, the GESI concept encompasses more than the equal rights, responsibilities, and opportunities for all genders but includes considerations of the needs and perspectives of young people, the elderly, people living with disabilities (PWD), ethnic minorities and nomadic communities, religious minorities, refugees, internally displaced people, remote-living people, the queer and intersex community, groups with less education, and low-income communities [38]. In this paper, these are referred to as people who are marginalised. The concept is integral to many SDGs, including SDG5 on 'gender equality' and SDG16 on 'inclusive societies'. With the SDGs calling to leave no one behind, GESI acts as a prerequisite to achieve all other SDGs. It is vital to also recognise that aspects of marginalisation may intersect, for example when a woman is disabled or elderly, or both [21]. This intersectionality implies that every person embodies different social characteristics regarding their gender, race, sexuality, which do not operate in silos but are interdependent and affect a person's life in their combination [7,39].

In the energy sector, GESI is important for the process of creating equitable and inclusive energy systems across the world. Every person's way of accessing and using energy is individual, which creates a diversity of needs that should be considered in planning [40]. The concept of GESI is often framed as a continuum comprising different 'tiers' (see Fig. 1), which indicate the degree to which gender equality and social inclusion are acknowledged and integrated into policies, programmes and projects [41]. GESI-neutral approaches are seen to ignore existing inequalities, assuming interventions affect everyone equally, often reinforcing exclusion. For example, electrification projects that do not consult affected communities in their planning may overlook access barriers faced by marginalised groups. GESI-aware or -sensitive approaches recognise inequalities but do not actively respond to or address them. For instance, a clean cooking initiative may acknowledge gender imbalances but not design the financing models or ownership structure to enable women to participate on an equal footing. GESI-responsive initiatives intentionally adapt to the different needs of marginalised groups to reduce disparities, such as ensuring that information and technologies on energy solutions are accessible to people with disabilities and information campaigns reach all people in rural areas as well as urban areas. At the most advanced end, GESI-transformative approaches seek to challenge and change the underlying power structures and norms that produce inequality. This may entail energy policy reforms that mandate representation of marginalised groups in energy planning and target structural barriers to energy access.

In many LMICs, GESI in energy has gained attention among researchers and international networks, such as ENERGIA, over the last 30 years [42]. Despite these efforts and the widespread recognition of the importance of GESI importance for just energy transitions, the uptake of GESI-responsive policies remains slow [43,44], as does their implementation [35]. Given this context, this paper adopts the GESI-responsive tier as the primary point of analysis along the GESI continuum. While impact begins at the GESI-sensitive tier where approaches may include consultation processes and an awareness of structures of marginalisation, such approaches may not result in structural or sustained transformation. GESI-transformative approaches, however, aim to challenge power dynamics and the roots of inequality. Between these two tiers, the GESI-responsive tier represents the minimum threshold at which positive impact can be meaningfully achieved by enabling inclusive participation and decision-making. It is therefore a pragmatic and realistic benchmark, especially considering the current stage of policy and institutional development in the energy sectors of many LMICs.

2.1. Overview of energy planning in Kenya

The Kenyan Energy Policy, 2025–2034, recognises the importance of energy in development, with decisions on energy issues affecting all economic sectors [45]. Energy planning is the mandate of national level institutions, as well as the 47 county governments. At the national level, the main stakeholders include: the Ministry of Energy and Petroleum (MoEP), which is responsible for policy formulation for efficient management and expansion of the energy sector, and development of a long-term vision for the energy sector [46]; power generating company Kenya Electricity Generating Company PLC, which is 70 % government and 30 % privately-owned [47,48]; the fully government-owned transmission company KETRACO [49]; and the Kenya Power and Lighting Company

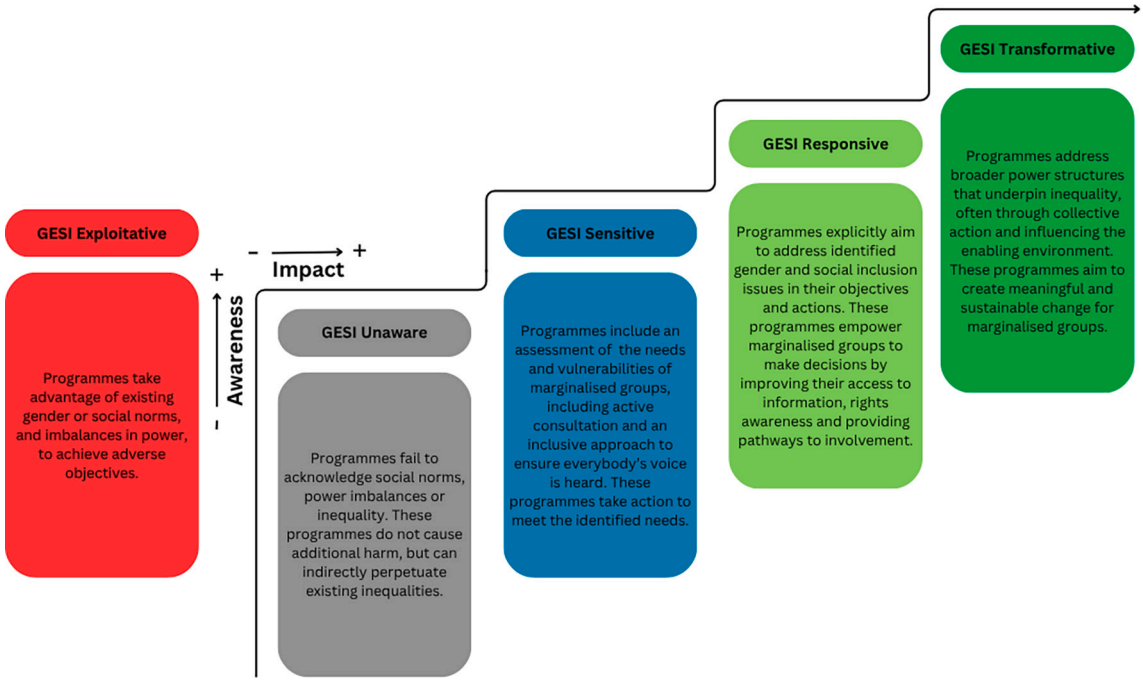

Fig. 1. The GESI Continuum.
Source: Hirmer, S. and Stockport, B. (2025) [41].

PLC, which is 50.1 % government-owned [50] and responsible for operating the majority of the electricity transmission and distribution system for onward transmission, distribution and supply to consumers [51]. While most of the energy generated in Kenya is controlled by KenGen, private sector actors, such as Independent Power Producers and Emergency Power Producers and Prosumers, also play a role [52,53]. The regulatory authority is the Energy and Petroleum Regulatory Authority [52]. The Rural Electrification and Renewable Energy Corporation focuses on supplying remote and underserved areas, as well as leading Kenya's renewable energy efforts [54].

The Constitution of Kenya 2010 devolved certain powers and responsibilities from the national government to the newly-established 47 county governments to allow for more localised approaches to planning [18]. The Energy Act of 2019 then set out the roles of the national and county governments with respect to energy infrastructure. It provided the national government with responsibility for policy formulation, energy regulation and licensing, as well as the operation and development of energy infrastructure [55]. The Act requires county governments to prepare their own County Energy Plans (CEPs) as part of the Integrated National Energy Planning (INEP) Framework [56]. However, minimal guidance has been provided to counties on the tools and planning approaches to assess county energy demand [57]. Due to their limited development, CEPs are not yet able to inform country-level energy planning from the bottom up [13]. Furthermore, a lack of gender-disaggregated data in county-level energy planning has been found [58].

Access to electricity has been an important element of Kenya's energy policy, and the country has made significant progress in providing a growing population with access to energy. Since 2000, electricity access has increased from just 15 % to 76 % in 2023 [59]. Challenges to further increasing electricity access in Kenya include high connection and supply costs, especially in rural and sparsely populated areas, inadequate incentives for private investment, and insufficient capacity to meet demand [8,23,60]. Additional barriers identified in the literature involve bureaucratic delays, corrupt practices, inefficient power transmission, unreliable supply and frequent outages [57,61–63].

2.2. GESI and energy in Kenya

The constitutional and legislative foundations for GESI were established by the 2010 Constitution, which articulated equality principles in Articles 10, 27, and 53–57 that mandate affirmative action for marginalised groups. Other legislative instruments followed, such as the Persons with Disabilities Act 2003 (amended 2023), which established protection frameworks subsequently referenced in energy access programs. National policy frameworks have demonstrated evolving sophistication in GESI approaches. The National Gender and Development Policy 2000 represented early gender mainstreaming efforts but remained largely gender focused. Its 2019 revision showed greater awareness of intersectionality, explicitly addressing women and girls, men and boys, PWDs, youth, older persons, and marginalised communities [64].

The Ministry of Energy's Gender Policy 2019 marked the energy sector's first comprehensive GESI framework. In its preface the Principal Secretary stated the intention of the policy "…*to strengthen institutional capacities for gender mainstreaming as a basis for gender-sensitive project formulation, implementation, and impact evaluation of all energy projects in Kenya*" (p.ii). While the policy defines gender equality broadly to include "*absence of discrimination based on a person's sex, age, social class, religion, colour, race, or political affiliation*" (p.40), implementation indicators focus primarily on women's participation metrics rather than broader social inclusion [19]. Nonetheless, the policy remains pioneering in efforts to mainstream gender across the energy sector.

Kenya continues to face challenges in delivering an equitable and inclusive energy system that meets the needs of the diverse population. Energy access and the burdens associated with energy collection and use have been found to vary across settlements and households. For example, women are disproportionately affected by cooking with traditional fuels, such as biomass, which has well documented negative impacts on health and time [21]. Inadequate access to energy among other services and resources is also recognised in the Kenya Women's Economic Empowerment Strategy 2020–2025 by the State Department of Gender as a strategic issue in gender equality, alongside women being affected by lower social standing and respect as well as unequal

economic opportunities [65]. Winther et al. [66] discuss how men typically have power over household equipment and finances, which may prevent women from purchasing and using the energy devices they need. In their study, they report positive effects of women's participation in power generation, such as being trained as solar engineers, on gender stereotypes in communities. The opportunity to sell solar systems and participate in power production and distribution also provides women with important sources of employment, as well as better lighting [17,67,68].

Stigma and discrimination against certain groups impacts marginalised people's access to services and resources on a structural level. For example, PWDs face stigma and exclusion, leading to lower employment opportunities and lacking access to services, such as public transportation, energy and health services [69–73]. Residents of informal settlements, who account for as much as 60 % of Nairobi's population, often lack formal access to the electricity grid, which means they rely on illegal electricity connections that may be unsafe, inefficient and unreliable [74]. Older people in Kenya may face several challenges, including lower mobility, healthcare needs, social isolation, and low, fixed incomes – typically limited pensions or family support – that shape their energy needs, but also awareness of what technologies and support are available to them. One consequence is that older people may struggle to afford safer, modern energy options and instead rely on traditional fuels, such as firewood and charcoal [75,76]. In humanitarian settings, energy access has been found to vary across settlements and households, with energy poverty causing conflicts and disproportionately affecting women [77–79].

3. Methods

The analysis draws on primary and secondary data to understand how GESI is considered in energy planning and the implications at the national and sub-national levels in Kenya. A review of Kenya's policy and planning landscape was undertaken, which covered relevant policies and legislation in energy and GESI. This review provided an understanding of policy ambition in this field in Kenya, and was followed by primary data collection in the form of semi-structured qualitative expert interviews.

3.1. Policy review

This study conducted a review of GESI policies related to energy planning in Kenya from 2000 to 2024. The year 2000 was selected as a starting date to capture Kenya's democratisation period and early gender mainstreaming efforts, while the endpoint reflected the most recent policy developments available.

Identification of relevant documents involved searching official government repositories using predetermined terms including "energy policy," "gender mainstreaming," "social inclusion," "gender monitoring," and "energy governance." Primary sources included the Ministry of Energy portal, county government websites, the National Gender and Equality Commission repository, and Kenya Law Reports database. The policies were reviewed with specific focus on:

- definitions of GESI, including how policies conceptualised GESI and intersectionality
- identification of affected groups, specifically which marginalised populations received explicit attention versus more general references
- outlined implementation strategies and their ways of establishing institutional responsibilities, including the mechanisms, responsibilities, and procedures were prescribed for implementation, and
- monitoring frameworks, such as how monitoring indicators, reporting mechanisms, and enforcement provisions were structured.

The policy review faced several limitations. Document availability was uneven, particularly for county-level policies and implementation guidelines that might have contained more detailed GESI provisions. Some key documents existed only in draft form or required formal information requests, which limited comprehensive assessment of policy intentions versus practical guidance. Based on the available documents, however, the policy review findings directly informed the design of expert interviews to assess how these policies work in practice and identify gaps in implementation.

3.2. Expert elicitation

This research draws on 23 qualitative semi-structured interviews with stakeholders in the fields of energy and GESI. Semi-structured interviews were used to elicit open-ended responses while ensuring consistency across interviews through a predefined set of questions with consistent phrasing. This was to build on and complement the literature review undertaken. Most interviews (19) were conducted between February and August 2022 and lasted between 40 and 120 min. They predominantly took place face-to-face, typically in the participant's place of work, although a number took place online via Zoom when it was not possible to travel to the interview (typically for county-level interviews). Participants were initially selected through purposive, or non-probability, sampling, wherein participants are identified and selected because of their knowledge about and experience on a phenomenon; in this instance, because they were experts working on energy at the intersection with planning, gender, equality, and/or social inclusion. This was followed by snowball sampling wherein additional participants were identified through referral by the initial participants. Participants worked for a range of organisations, including relevant government, private sector, international organisations, and non-governmental organisations. A total of 15 women, and eight men were interviewed for this research. In July and August 2024, a second round of interviews was conducted with four further experts to test the relevance of the main findings that emerged from the initial interviews two years previously. These were again semi-structured conversations and focused on what had happened since 2024 in GESI in energy planning and to assess whether the previous findings still held true. All follow up interviews were conducted online. An overview of the 23 participants based on their affiliations and in-text codes is provided in Table 1.

The interview guide (provided in the supplementary material) was co-developed by the research team, who are experts in energy access and planning and gender studies. Starting with the study's research aims, the team developed interview questions that would address the broad areas of interest and findings from the policy review. Care was taken to use open questions that would encourage complete and honest answers, and to ensure a logical flow to the conversation. Once drafted, the guide was reviewed by two colleagues with expertise in GESI and energy, revised and piloted with two participants to ensure clarity and suitability of the questions. No revisions were made to the guide following this pilot. The interviews centred around four broad areas of discussion:

- Understandings and definitions of GESI
- Consideration of GESI in energy planning
- Factors that impact the ability of socially excluded groups to access energy services and technologies
- Recommendations for more inclusive energy planning and provision.

All interviews were recorded with permission of the participants, transcribed and then analysed using the software NVivo. Thematic analysis was used to analyse the interview data, which involved familiarisation with the data, generation of initial codes, identification of themes, and subsequent review of those themes. Initially, a deductive approach was used whereby code categories were identified that broadly corresponded with the questions and broad areas of discussion during interviews [80,81]. This was then followed by an inductive

Table 1
List of participants interviewed for this research.

Organisation		Work Focus	In-text Abbreviations	Gender of participant	Year interviewed
National government	Ministry of Energy and Petroleum (MoEP)	Gender, development	NG1	Female	2022
		County energy planning, GESI	NG2	Male	2024
	National Gender & Equality Commission (NGEC)	Gender, social inclusion	NG3	Female	2022
	Ministry of Gender, Culture, the Arts and Heritage (MoG)[a]	Gender	NG4	Male	2022
County government	Homabay	Gender	CG1	Female	2022
	Kisumu	Energy	CG2	Female	2022
		Gender	CG3	Female	2022
		Energy	CG4	Male	2022
	Kitui	Environment, Energy	CG5	Male	2022
		Energy	CG6	Female	2022
		Gender	CG7	Female	2022
	Kilifi	Energy	CG8	Male	2024
Donor		Energy, Transport, Humanitarian Settings	D1	Female	2022
		Energy, Development	D2	Male	2022
International Non-governmental Organisations (INGO)		Energy, Climate Change	INGO1	Female	2022
		Energy, Transport, Climate Change	INGO2	Female	2022
		Gender, Social Inclusion	INGO3	Female	2022
Civil Society Organisations (CSO)		Energy, Gender	CSO1	Male	2022
		Energy, Gender	CSO2	Female	2022
		Disabilities Advocacy	CSO3	Female	2024
Private Sector (PS)		Manufacturing, Communication, Gender	PS1	Female	2022
		Energy, Mobility, Social Inclusion	PS2	Female	2022
Academia (AC)		County Energy Planning Technical Assistance	AC1	Male	2024

[a] Formerly the Ministry for Public Service, Gender, Senior Citizens Affairs & Special Programs.

coding process whereby new codes were identified in line with the data generated during the interviews [80,81]. Through this process, four main themes were identified, along which the analysis is structured:

- Understandings and conceptions of GESI and gender
- Marginalisation in energy - which groups are currently left out in energy planning and why
- Current status of GESI and energy national and county policy and planning
- Recommendations for GESI inclusive energy planning in Kenya.

3.3. Limitations and future research

This study was exploratory and interpretive by nature. As a result, the findings do not offer conclusive answers to the questions posed about GESI in energy planning in Kenya, rather they are indicative of the current status, opportunities and challenges. The participants interviewed as part of this study do not represent all those relevant to GESI and energy planning in Kenya. Furthermore, the participating energy decision-makers cannot represent all marginalised groups in Kenya. Future research efforts can be targeted towards interviewing or researching marginalised people who do not work in energy planning. They may name further barriers and enablers that the participants of this research did not discuss or perhaps identify. The sample might be affected by selection bias, as people who agreed to take part in this research may have higher interest in the integration of GESI in energy planning, and thus be more familiar with the topic.

3.4. Ethics and risk assessment

A risk and ethics assessment was completed and approved by the UCL Bartlett School of Energy, Environment and Resources' low risk ethics committee. All participants were asked for and gave their consent to participation in this research prior to the interview. To protect the participants' identities, all statements were anonymised. All personal information was kept strictly confidential.

4. Results and discussion

This section presents the analysis of the policy review and semi-structured interviews. It first provides the results of the policy review to examine the extent to which GESI aspects are currently integrated in energy policy and planning in Kenya. It then presents findings from the expert interviews in dialogue with existing literature to explore how stakeholder perceptions of GESI influence broader structural framings within policy and planning in Kenya's energy sector and thereby shape outcomes at both national and county levels.

4.1. Policies and programmes at the intersection of GESI and energy in Kenya

Kenya has enacted legislation which aims to protect the rights and ensure the welfare of specific marginalised groups. An overview of key policies and programmes relevant to GESI in energy policy and planning in Kenya is provided in Table 2.

Policies include the 2003 'Persons with Disabilities Act' [82], following which the National Council for Persons with Disability was established [83]. In the Kenyan energy sector, governmental actors, civil society, development partners and gender advocates began working together to mainstream gender in the mid-2000s, and began to conduct gender audits of Kenya's energy policies and projects in 2006 [84]. In the same year, Vision 2030 - Kenya's development strategy for the country to become a middle-income economy by 2030 - recognised *'the equal rights of men and women to enjoy a high quality of life and to benefit from equal opportunities'* [84]. The strategy also included a target year for universal power access, which was first set to 2030, then changed to 2022 in 2013, and has since been revised back to 2030 [60].

In 2010, principles of equality and non-discrimination were enshrined in the Constitution of Kenya [18]. Article 27 guarantees the right to equality and freedom from discrimination, while Article 10 emphasises social justice, human dignity, equality, inclusiveness, non-discrimination, and protection of marginalised groups. State organs and public officers are obligated to respond to the needs of marginalised

Table 2
Overview of policies and programmes relevant to GESI in Kenya since 2000.

Year	Policy / Programme / Institution	Description
2000	National Gender & Development Policy (first edition)	Kenya's first national gender, policy aiming to mainstream gender across sectors
2003	Persons with Disabilities Act	Legal framework to protect the rights of PWDs; established the National Council for Persons with Disabilities.
2006	Kenya Vision 2030	National development strategy that defines development priorities until 2030.
2010	Constitution of Kenya	The Constitution is the supreme legal instrument that establishes rights for all Kenyans, with some provisions protecting marginalised groups from previous marginalisation. Enshrined principles of equality, non-discrimination, and affirmative action for marginalised groups.
2011	National Gender and Equality Commission (NGEC) Act	Established a dedicated commission to promote gender equality and freedom from discrimination in accordance with Article 27 of the Constitution.
2011	National Gender and Equality Commission (NGEC)	Oversight body to promote equality and monitor discrimination across all social groups.
2011	Kenya Informal Settlements Improvement Project (KISIP)	Urban infrastructure upgrades (incl. lighting, electricity) in informal settlements.
2013–2024	Energy Access Expansion (e.g., Last Mile, KOSAP, RE Programme)	Series of major projects to improve grid and off-grid energy access in low-income, rural, and underserved areas.
2014	National Policy on Older Persons & Ageing 2014	Social protection framework addressing the unique challenges of older persons
2017	Kenya Comprehensive Refugee Response Framework	Framework for transitioning from a camp-centred approach to community integration.
2018	Kenya National Electrification Strategy	Sector strategy setting roadmap to universal electricity access, grid and off-grid, targeting remote or underserved populations
2019	National Energy Policy	National energy policy that, in its text and annexes, recognises GESI issues
2019	Energy Act 2019	Consolidates energy laws and clarifies national & county roles for energy delivery. The Act establishes institutional arrangements (ministries, REREC, regulators) that are responsible for implementing electrification and must comply with constitutional equality obligations — making it important when assessing whether energy institutions fulfil GESI obligations.
2019	National Gender and Development Policy (Second edition)	Requires gender mainstreaming in national and county development plans, budgets, and policies.
2020	Ministry of Energy Gender Policy	Framework for integrating gender in energy planning and programming.
2020	National Energy Efficiency & Conservation Strategy	Provides a roadmap for enhancing energy productivity across key sectors. Promotes several measures to address the differentiated impacts of energy use on different social groups.
2021	Energy Sector Gender Committee	Multistakeholder group guiding gender policy implementation in the energy sector.
2021	Refugees Act 2021	Refugee management framework
2023	Persons with Disabilities Act (amended)	Framework that protects the rights of persons with disabilities and ensures their full inclusion in society.
2024	Persons with Disabilities National Policy	Addresses structural barriers for PWDs, including intersectional needs across gender, age, and poverty lines.
2025	revised National Energy Policy drafts 2024/25	Provides a forward-looking roadmap for Kenya's energy sector. Highlights the goal of building an inclusive, resilient, and sustainable energy system that leaves no Kenyan behind signalling strong alignment with GESI principles.

groups as per Article 21(3), while Article 56 establishes that the state should pursue affirmative action for minorities and marginalised groups in governance processes, education, employment and access to services. Articles 53, 54, 55, and 57 specifically stipulate rights, entitlements and affirmative action for children, PWD, youth and the elderly respectively. Following the Constitution, the National Gender and Equality Commission was established in the NGEC Act in 2011 to promote gender equality and freedom from discrimination for all persons in Kenya [85]. In the same year, the Kenya Informal Settlements Improvement Project was started which focuses on upgrading infrastructure, including lighting and electricity access, and improving living conditions in urban informal settlements [86–88].

Regarding energy access, Kenya saw a steep rise in the electrification rate especially between 2013 (32 %) and 2019 (75 %) [59], which can be attributed to a list of still ongoing programmes and initiatives for low-income households, informal settlements, and remote areas, including:

- the Last Mile Connectivity Project, which subsidises grid connection costs for parts of the population in low-income, rural areas which are hard to reach [89,90]
- the Rural Electrification Programme which aims to improve electricity access in public facilities in rural areas [91]
- the Kenya Off-Grid Solar Access Project (KOSAP), which addresses power shortages in specific under-supplied, less-central counties [92]

These programmes significantly improved access for low-income households in rural communities, low-income urban areas, and other settlements which had not previously been served [19]. The MoEP itself, however, points out that many of these energy programmes "*are gender neutral [and] therefore fail to address the special needs of the poor especially female headed households, widows, widowers and the elderly without a pension*" (p15) [19]. In 2019, the State Department for Gender Affairs, part of the Ministry of Public Service, Youth and Gender, published a National Policy on Gender and Development, building on a Gender Policy first developed in 2000 [93]. It provides that all planning, budgeting, and implementation of development programmes incorporate a gender lens both at the national and county level [94]. In 2020, the MoEP launched a gender policy for the energy sector, which sought to provide a framework for state and non-state actors to integrate gender considerations in energy projects, budgeting, programmes, planning and policies [19]. The policy defines the term 'gender equality' as the 'absence of discrimination based on a person's sex, age, social class, religion, colour, race or political affiliation' (p.40), hence including other aspects of marginalisation besides gender. The policy also discusses the importance of considering poor rural and urban poor areas, PWDs, and youth in energy planning, recognising that these groups have different levels of energy access with policies affecting them differently.

However, the commitments and strategies outlaid in the policy mostly revolve around the equality of women and men in energy policies and programmes, and the targets and indicators proposed for monitoring and evaluation similarly focus on the representation of women and men at the MoEP and in the STEM sector, implementation of the two-third rule,[1] the establishing of breastfeeding facilities, and the number of women and men trained on gender equality and energy. Further marginalising factors such as age, disability, remoteness and poverty are mentioned to a lesser extent.

Further developments include the establishment of an energy sector gender committee in 2021 to guide policy implementation. This Committee consists of representatives of various organisations including semi-autonomous government agencies in energy, civil society, and Kenya's Council of Governors [85]. The NGEC also introduced a gender mainstreaming reporting survey to assess the progress of gender mainstreaming initiatives within governmental authorities [85]. For the financial year 2022 to 2023, different energy companies received high scores, with the MoEP achieving a score of 97/100, Kenya Power 93/100, KETRACO 97/100, and KenGEN 95/100. In 2024, the Ministry of Labour and Social Protection also published the 'Persons with Disabilities National Policy' which discusses ways to address critical barriers to the upholding and protection of the rights and dignity of PWDs [96]. It also addresses intersectional needs, such as the specific challenges for children, youth, women, and older persons with disabilities.

These efforts demonstrate Kenya's policy commitment to enhancing inclusivity and addressing the needs of different groups in the area of energy access. However, policies do not necessarily translate into practical implementation, leaving scope to strengthen inclusive energy planning and provision.

4.2. Expert perceptions of GESI in energy planning

The interviews with the 23 experts revealed how the decision-makers' perceptions of and experiences with GESI have impacted energy policy and planning in Kenya. Key themes included: understandings of GESI; marginalisation in energy planning; implementation of GESI in energy policy and planning; and recommendations for improving GESI integration in energy planning. These issues are elaborated upon in the following sections.

4.2.1. Understanding GESI: misconceptions, narrow framings, and emerging nuance

While GESI is intended to promote inclusive development for all marginalised groups, interviews revealed varied and sometimes conflicting interpretations of the concept. These ranged from narrow associations with gender alone to more holistic understandings that integrated intersecting dimensions of social inclusion.

A key finding from the interviews is that GESI is commonly understood in a narrow sense, with many participants equating the concept primarily with gender equality—particularly inclusion of women—while overlooking broader dimensions of social inclusion and intersectionality. When asked to define GESI, nearly half of respondents focused on equal rights and opportunities for men and women, stressing the *gender* part of GESI, while failing to mention other marginalised groups or their intersection with gender. This narrow interpretation was observed across all categories of organisations represented in this research (see Table 2).

Participants noted that gender-related issues were often addressed before other intersecting aspects of marginalisation.

CSO3 argued that "*when it comes to GESI, women are always (…) the biggest beneficiaries*." This participant pointed out that even within programmes such as the Access to Government Procurement Opportunities programme, which legally prescribes that 30 % of government procurement opportunities are dedicated to enterprises owned by women, youth, and PWDs, women were the majority recipients. This prioritisation was also recognised in the MoEP's Gender Policy which noted that the uptake by rural women, youth and PWDs was limited [19], suggesting that intersectionality and the varying experiences of marginalisation were not adequately accounted for. While the fact that women are the main beneficiaries is not in itself problematic, the concern lies in which women are primarily able to access these opportunities.

This narrow focus on gender has been identified in GESI studies in other countries and sectors. For instance, in energy planning in Nepal, GESI policies have been shown to place emphasis on women while neglecting social inclusion [34,35]. An examination of GESI in WASH interventions in East and Southern Africa [32] found that most interventions prioritised women, with comparatively little attention given to other groups or intersectional factors, which is echoed in a review of WASH interventions in LMICs more generally [27].

Despite the tendency of some participants to narrowly frame GESI around gender, othres demonstrated a broader, more inclusive interpretation of the concept. AC1, for example, stated that, in their experience in county energy planning, their approach was *"equal to all*" and went *"beyond the binary"*. This was supported by NG2 who highlighted that there were some programmes specifically *"geared towards social inclusion",* such as KOSAP or electricity interventions in informal settlements which targeted low-income or remote communities. In some instances, 'gender' was used more broadly and served as an umbrella term that included all marginalised groups. For example, NG4 described that the remit of the gender department not only included the needs of women, but also of PWDs, the elderly, and other marginalised groups. This reflects the definition of 'gender equality' in the MoEP Gender Policy (see Section 4.1), which refers not only to men and women, but also to other marginalised groups. Here, 'gender' is perceived to be inclusive of other marginalised groups, although the policy's actual commitments, strategies, and indicators for monitoring progress, focus more on women than on other marginalised groups.

4.2.1.1. Scepticism and misperceptions about GESI. Beyond these perceptions, some participants explained that they had encountered scepticism and resistance to GESI, with some people arguing that the objective of GESI was to empower women to the disadvantage of men:

> *"in any country, this gender thing is overrated. It tends to favour women. It pits men against women (…) Worldwide, not everyone wants to empower women"* (CG2).

CSO1 highlighted that, particularly at the county level, women who speak up might be seen as a *"threat to thousands of men who are around you",* while PS2 caveated that women should not be empowered at the expense of men, as men would consequently not know how to engage with women.

These statements reflect the misunderstanding of the GESI concept, which seeks to promote equal opportunities for marginalised groups without diminishing the rights of non-marginalised groups. However, one participant (CG7) challenged these views, emphasising that affirmative action in support of women does not imply that men's needs would be neglected:

> "*We are fighting this war of inclusion and equality and we lean more towards women. Some people would say that we are forgetting the boy child, but the boy child is yet up there."*

Some participants argued that while scepticism and resistance to GESI by some people persists, continued dialogue and education would help to shift perceptions towards a more balanced understanding of GESI's role in promoting equality without disadvantaging any particular

[1] As per Art 27(8) of the Constitution, no more than two thirds of staff shall be of the same gender in appointive or elective bodies, which includes national and county governments [18]. However, the bill that seeks to legislate this constitutional requirement has not been enacted [95].

group. For example, this may include increasing men's awareness of the role that women can play in the economy and motivating them to support women's ownership of land and resources [28].

4.2.1.2. Marginalisation in energy planning: impacts and important factors. The interviews explored whether and why participants thought that women and girls, PWDs, refugees, ethnic minorities, the youth, IDPs and the elderly were or were not adequately considered in energy planning at national and county-level. Perceptions of temporariness, political interference, data gaps and structural barriers, such as poverty, location, cultural norms and lack of information emerged as important issues. Understanding the specific energy needs, uses, and constraints faced by different marginalised groups is essential for planning energy systems that effectively respond to their realities and support equitable access. Alongside the specific needs, below we also discuss the structural inequalities and stigmatism that currently exist for various groups.

Many participants highlighted that **refugees and IDPs** were excluded from planning, explaining that their living situation was perceived as '*temporary*' (CG2) and therefore not adequately accounted for in national or county level planning (PS2). This was exacerbated by the lack of clarity about who was responsible for planning for refugees' energy needs. Some participants pointed to development partners, such as the United Nations, while others mentioned the national or county governments.

Regarding **women and girls,** interviewees discussed missed economic and educational opportunities resulting from inadequate consideration of their needs in energy planning. For instance, in some girls' high schools, students tasked with collecting firewood – an energy burden not placed on boys – highlighting a lack of planning for gender equitable energy provision in school infrastructure (INGO2). NG4 reflected on how **girls'** lower educational attainment reduced their ability to pay for energy services and thereby limiting income-earning potential:

> "*Low education of girls from primary transiting to secondary school to tertiary colleges and university, denies them self-employment or formal employment that would earn them adequate income to boost their affordability of energy solutions*."

Stigma around **older persons** was highlighted by NG4 as a barrier to equitable energy planning. They noted that older people were often perceived to not need modern energy services and technologies:

> *"The elderly are assumed to be people who belong to darkness. (…), so they are never thought about yet they are the ones that need energy access the most because their seeing capacity is impaired, their ability to cook is impaired, and they need better technology for faster and more efficient cooking solutions."*

Ethnic minorities were also reportedly underserved (INGO2, CSO2, NG3, PS1), with some groups argued to have no access to government services (PS2). Until recently, energy providers had focused on formal urban areas, neglecting **informal settlements and low-income areas**, such as Turkana, Samburu, West Pokot, Baringo, Kajiado and parts of the Coast.

CSO3 described how for **PWDs**, there was no form of social protection or programmes that supported them during periods of unemployment. This economic disadvantage, arguably, then impairs their ability to pay for energy.

Marginalised groups not only encounter stigma and discrimination, but their specific social and economic circumstances may be inappropriately considered in energy planning. This perpetuates inequalities as it prevents those impacted from taking part in energy-enabled economic activities. For example, high connection fees and tariffs may prevent people from accessing energy services (NG4) or low income households may need to spend their limited resources on food, school or hospital fees (INGO2, PS2), and opt for firewood or charcoal over cleaner cooking solutions, such as electric cookers, or those requiring the purchase of gas, briquettes or pellets (CG4, INGO2). Geographic barriers in so-called 'hard to reach' areas may perpetuate this as energy services and technologies in such areas are likely to be higher. While some initiatives seek to address these economic barriers, e.g. KOSAP and the Rural Electrification Programme (NG1, NG4), challenges persist.

4.2.2. Implementation of GESI in energy policy and planning

Perspectives on whether GESI had been fully institutionalised varied among participants, but a common theme emerged regarding the translation of policies into action at the county level. This was described as varying significantly between counties, often inadequate and inconsistent, which was largely attributed to a lack of capacity and funding.

4.2.2.1. Drivers and gaps in institutionalising GESI in energy planning. Perceptions of what motivated decision-makers to include GESI considerations in their planning processes, as well as how this affected the extent to which GESI was put into practice, varied. Several participants highlighted that the rights of all groups and principles of equality and non-discrimination were explicitly enshrined in laws and policies, most importantly the 2010 constitution (CG2, CG4, CG5, NG1, NG2, NG3, NG4), as well as the MoEP Gender policy (INGO2, INGO3). However, several interviewees thought that GESI had yet to be institutionalised properly (CG7, D1, PS2, CSO2) and that GESI was largely driven by development partners. Some felt that national and the county governments saw GESI as "*ticking the box*" (PS2, CSO2, AC1) and argued that, in some instances, GESI considerations were only included in planning and projects when demanded by development partners (INGO2, CSO1, CG3). This perception of GESI being driven by external development partners rather than by local actors mirrors findings from other countries, where GESI-inclusive energy policies were found to be developed mostly in individual initiatives as conditions set out by development partners [35].

4.2.2.2. Variation in policy implementation across counties. Despite the strong promotion of inclusiveness and equality in Kenya's constitution and related policy frameworks (see Table 2), the operationalisation of GESI policies and their operationalisation remains challenging. CSO3 noted that *"we are still a long way to go in ensuring that the policies that we have in place are implemented"*. The implementation varied significantly across counties, being slow in many (INGO2, INGO3), which participants attributed to a range of factors, including political will, differing priorities and interest in GESI more broadly. CSO3 noted their organisation's collaborative relationship with Nairobi County government for knowledge sharing and participatory planning of PWDs, while acknowledging that more work is needed to establish this practice across all counties.

Although the INEP has devolved energy planning to Kenya's 47 counties to ensure that planning corresponds to local needs, a key barrier mentioned by participants was that interest in and capacity to develop county-level energy plans varied greatly across the country. For example, Kilifi county has a comparatively advanced energy plan, which was actively informing the implementation of energy measures (NG2), while other counties lack a dedicated energy department or directorate. In these latter counties, energy was "*bundled in a ministry with a lot of other [topic areas]"* (INGO2), an issue also raised by NG1, NG2, and CG7. Where progress is slow, conversations about the importance of energy and GESI remain difficult, as outlined by INGO3:

> *"Even recognition of the important roles that energy plays has been very low, so we are not just talking about gender"*.

This was evident in other sectors. For instance, a 2021 review of Kenya's 2012 gender policy for agriculture found that its implementation differed significantly across the country due to a lack of human and financial capacity at county-level, leaving systemic gender inequalities

largely unaddressed [32]. Arguably, the devolution of governmental responsibilities via the INEP framework, is a step in the right direction as counties are better placed to identify and decide where energy access is needed and how to supply it. However, the necessary resources need to be in place to support this devolution. An initiative which could partially address the imbalances evident between counties is the Equalisation Fund, which, as per Article 204 of the Constitution, aims to provide finance to address inequalities between the counties in basic services, such as energy.

4.2.2.3. Barriers to GESI policy implementation. Several factors inhibit effective execution of GESI policies, including limited institutional capacity, poor guidance on GESI operationalisation, and weak interdepartmental coordination, lack of knowledge sharing, gender imbalance in staffing, insufficient budget allocation, and corruption. Some of these were also cited in a study on the devolution of energy planning in Kenya [97], and were described by one interviewee (PS1) as follows:

> *"I do not think the issue is about taking it seriously, it is the implementation part of it because we find that there is a lot of policies developed to push the youth, women, and PWDs (…), but the implementation is where the road should meet the rubber. But that is where the issues come in, so you find maybe there [are] no budgetary provisions for the same (…). Or maybe even they do not have the capacity to drive the same at the county level".*

NG2 noted that energy planners and project developers have limited capacity to effectively operationalise GESI policies. This results from understaffing, loss of expertise through frequent reshuffles of government officials at county-level (INGO2), and lack of guidance and training. A lack of expertise on GESI in energy was also cited as a barrier, which was attributed to a wider lack of GESI experts in Kenya (NG1). This is similar to findings elsewhere where a lack of training, sensitisation and awareness of GESI were cited as key barriers to mainstreaming GESI in planning processes [28,29].

Weak interdepartmental coordination and lack of knowledge sharing were also cited as challenges for progress. For example, CG3 and CG4 highlighted challenges regarding coordination, communication, and knowledge sharing between different ministries and departments, leading to a lack of awareness of relevant initiatives and knowledge products, involvement of relevant teams in implementing measures across counties, and input or support from relevant stakeholders. To elaborate, an energy project might be planned and implemented by either the national government, the county government, or private sector without involvement or coordination between these parties (CG4). It is noteworthy that the INEP Framework, especially with the newly integrated GESI component, has improved coordination (NG1) with the hope for better knowledge sharing across counties (CG8).

Participants reported that gender imbalances in staffing hindered GESI sensitivity in energy institutions. CG3, for example, argued that male-dominated environments in energy-related institutions lessens awareness of the importance of GESI. While Article 27(8) of the Constitution stipulates that no more than two thirds of staff should be of the same gender in appointive or elective bodies – which applies to county governments – this provision has not been translated into practice. Fifteen years since the Constitution was adopted, the Parliament has yet to pass the legislation required to enforce this article, despite multiple court rulings and public commitments [97].

Insufficient budget allocation was another barrier mentioned. While connected to the lack of awareness and prioritisation, participants argued that GESI policies were not provided with adequate funding - especially at the county level (NG1, NG3, CG1, CG3, CG7, CSO1, INGO1, PS1). As a result, planners would "*find there is no one interested in a document that does not have a budget*" (CSO1). National ministries, such as the MoEP, had more funding and experience in integrating GESI in energy planning, whereas county government struggled to institutionalise GESI in their energy programmes and policies. The lack of finance slowed and hindered the development and implementation of policies (CG1).

Some participants raised concerns about political interference in energy planning decisions, explaining that energy was sometimes used as a tool during political campaigning. At the county-level, interviewees (CG1, CG3, CG7, PS1) also discussed corruption as a factor which undermined policy implementation. In one county, a Member of County Assembly allegedly demanded a bribe in order to pass a policy on Sexual and Gender Based Violence (CG1). The lack of funding for assemblies was cited as a contributing factor.

4.2.3. Participant recommendations for GESI inclusive energy planning in Kenya

This section sets out the recommendations from participants for advancing inclusive energy planning. Key recommendations included: promoting participatory and inclusive planning processes; moving beyond representation; increasing awareness and building capacity; and strengthening policy and legal frameworks related to GESI.

4.2.3.1. Promoting participatory and inclusive planning processes. The adoption of participatory approaches in energy planning were suggested to increase inclusivity in decision-making (NG1, CG4, CG5, INGO1, INGO2, PS1, PS2). Indeed, participatory planning is enshrined in the Constitution of Kenya and is also embedded in the INEP framework. CG8 – based in a county that NG2 described as comparatively more advanced in energy planning – reported that their county energy plan was developed drawing on participatory planning to ensure "*everyone is included*". This was echoed by AC1 who had supported energy planning processes in two other counties.

Some participants argued that more inclusive planning processes would need to consider the services that energy provides. In other words, planning should factor in the ways in which people use energy and what they use it for (D1). In this regard, AC1 reported that in Narok county, consultations had sought to understand what people needed energy for, which had led to a focus on ways to provide water to farmers for agricultural purposes. However, a thorough understanding of people's energy practices and especially the intersectional aspects of marginalisation - the "*layered forms of challenges*" (CSO3) - appeared not to be the norm. Indeed, this approach is comparatively new in the energy sector, as energy planning is typically approached with a focus on infrastructure and technology [98,99].

4.2.3.2. Moving beyond representation to address the root causes. Participants suggested that energy planning processes at all levels needed to go beyond merely securing representation by marginalised groups [see also [98]]. As one participant argued:

> "*A lot of times people (…) think gender is just ensuring that there are women and men participating in something, but essentially that is checking gender on the side and not really integrating gender in your work*" (D1)

Focusing on representation alone risks neglecting the root causes [34]. This issue extends beyond the energy sector and reflects a broader problem in policy and political processes where quotas are used as a proxy for meaningful representation, often resulting in tokenistic rather than genuine inclusion [35]. Those affected should be able to take part in decision-making, as emphasised in a GESI assessment of an agricultural and forest governance initiatives in Tanzania [29]. This requires better sensitisation of marginalised groups on the role of energy and enhancing information dissemination through appropriate mediums to empower people to be able to articulate their energy needs and participate meaningfully in consultation processes. This lack of voice and genuine participation reinforces exclusion and leads to energy plans that fail to reflect the realities and priorities of marginalised groups.

Here, CSO3 stressed the importance of working with dedicated organisations to ensure that diverse views and needs are acknowledged in planning:

> *"The important thing is to ask. And the best way to do that is to interact with an organisation (…) like us (…) to [create] more awareness of what GESI is, and entrenching the culture of multi stakeholder engagement because everybody holds the piece of a puzzle."*

4.2.3.3. Increasing awareness and building capacity. Participants also emphasised the need to increase awareness of GESI – beyond gender – and to build capacity at all levels of government and across sectors. With regard intersectionality and gender, CSO3 elaborated that there were:

> "*Issues to do with the rural urban divides, even if I live in an urban area, (…) a peri urban area, (…) an informal settlement? These intersectional identities provide extra challenges for women. (…) There's women with disabilities, young women with disabilities*".

Such considerations are rarely recognised in energy planning, unless specific advocacy organisations are included (CSO3). Thus, an important action point to emerge from the interviews was to increase education and awareness around GESI and energy. According to participants, this needs to focus on consumers, e.g. regarding the availability, functioning of and access to services, tools and solutions (CG6, CG7, CG8, INGO3, NG3, PS1), as well as capacity building of government staff and decision makers on GESI (NG1, NG4, CG1, CG4, CG7, INGO1, INGO2, INGO3). Capacity building was described as a continuous process of training, monitoring, evaluation and learning (INGO1, CG1). However, CSO3 caveated that awareness building had to be done via targeted communication, i.e. "*think[ing] outside the box*", especially to reach marginalised groups such as PWDs. The need to build capacity and awareness within institutions, policymakers and project stakeholders is again reflected in literature on other country contexts, including Bangladesh, Nepal, and Tanzania [25,29,35].

4.2.3.4. Strengthening policy and legal frameworks. Finally, participants stressed that the implementation of GESI in energy planning depended on strong political support, and emphasised that a strong policy framework was equally important for accessing financing:

Particularly at the county level, a strong legal framework was regarded as a crucial starting point (CG1). However, even if policies were in place, this did not mean that their provisions would translate into action, for example due to a lack of awareness on inclusivity as described by CSO3:

> "*Policymakers are not always inclusive. Just because we have good laws in Kenya about inclusion about rights of people with disabilities doesn't mean that policy makers know*".

This was echoed by NG2, who noted that:

> *"Sometimes something can be there already in the Constitution, but the awareness [is missing], and (…) sometimes people don't understand gender equality and social inclusion"*.

AC1 also raised the issue that there was no '*reward or punishment*' for following (or not) the inclusivity requirements of the INEP and suggested this needed to change to ensure enforcement of such frameworks. Furthermore, participants pointed out that some parts of the Constitution had not yet been implemented at the national level, leading county staff to feel less obligated to meet those requirements. NG2 suggested that a GESI framework for county officials could help to implement national level policies at the county level; this has since been added to the INEP framework. A framework or system for monitoring and evaluating GESI policy and its implementation is suggested as a starting point for transparency and accountability, and an incentive for implementation [25,34]. For this, the National Gender & Equality Commission's tool for evaluation of ministerial policies was seen as a step in the right direction.

5. Conclusion and recommendations

Achieving more inclusive energy systems is not only a policy issue, but a fundamental requirement for sustainable development. Kenya has made significant progress in increasing energy access by targeting and supporting remote and low-income areas, and in mainstreaming gender into policies and programmes aimed at increasing access to energy services. However, drawing on a review of relevant policy and literature at the intersection of GESI and energy planning, alongside 23 expert interviews examining how stakeholder perceptions of GESI shape policy and planning within Kenya's energy sector, this study highlights that substantial potential remains for creating an energy system that accounts for the needs of Kenya's diverse social groups. Specifically, this paper found that while progress has been made in recognising the needs of women and girls in energy planning, the needs of other vulnerable and marginalised groups remain comparatively underrepresented. Perceptions of the concept of GESI vary, with some interpreting it narrowly as focused on gender, while others raise concerns that GESI initiatives would seek to prioritise women at the expense of men.

Despite strong commitments to equality and inclusion through the 2010 Constitution, the MoEP Gender Policy and the INEP framework, there remains scope for improvement in their implementation. Particularly at the county-level, interviewees attributed the difficulty of realising equitable energy provision to various factors, including limited capacity and finance, as well as issues of tokenism and discrimination.

Further, the research revealed that GESI expertise was concentrated among a small number of individuals. As a result, the need for awareness and capacity strengthening to ensure that decision-makers are equipped to implement GESI policies effectively at all levels of governance emerged as a key issue. For this to be realised, there is a need for a dedicated budget, as well as improved collaboration and coordination between governments, civil society, and international partners.

More participatory approaches to policy development and implementation were recommended by participants, who deemed greater inclusivity essential. However, participatory planning needs to go beyond the fulfilling of GESI quotas to truly understand and address underlying structures of gendered and social inequalities that prevent marginalised and vulnerable people from accessing and using energy services. Furthermore, the research showed that planning must also consider the energy and wider development needs of different social groups; this call to go beyond technical approaches which largely focus on the provision of energy infrastructure alone has been made elsewhere [5,98,100]. It is increasingly accepted that while energy access is vital to deliver social and economic development, the provision of electricity or modern cooking fuels and technologies alone does not guarantee a just and inclusive energy system if interventions do not challenge structural barriers such as gender norms and unequal power relations. If such barriers are not addressed, patterns of power and marginalisation are likely to perpetuate, and the potential benefits of energy access for all will not be fully realised. Energy needs and practices must be understood in their diversity rather than addressing different societal groups in silos. This requires taking an intersectional approach to participatory and inclusive planning will not only benefit marginalised communities, but also contribute to the achievement of SDG7 and other key development goals in Kenya and beyond. Other countries that seek to integrate considerations of GESI in energy planning can benefit from Kenya's experiences, including the insights related to conceptions of GESI, the implementation of policies, and the design of transformative approaches that address power dynamics and inequalities.

Finally, and to conclude, while notable strides have been made in mainstreaming GESI into energy planning, especially at the national level, deeper integration is needed. As a regional leader in the delivery of access to modern, Kenya is well positioned to provide a powerful

example by embedding GESI comprehensively and consistently throughout its energy planning processes, thus ensuring that energy access delivers meaningful development for all.

CRediT authorship contribution statement

Janina Luisa Fuchs: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Lucille Akelo Onyango:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Stephanie Hirmer:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Julia Tomei:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

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Data availability

The data that has been used is confidential.

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